



6.7.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can subtract integers using a number line.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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2. I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
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Unit 6, Lesson 8: Integer Subtraction



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Target

- ☐ I can subtract integers with values greater than 100.



6.8.1 Warm-Up: Self-Reflection

For each statement, rate yourself on a scale of 1-5. A rating of 5 means “absolutely yes” and 1 means “absolutely not.”

Statement	Rating
I can subtract integers using integer chips.	
I understand how to show subtraction of integers on the number line.	
I can subtract integers without using integer chips or a number line.	
I can explain to someone else how to subtract integers.	



6.8.2 Guided Practice: Subtracting Integers

1. Two expressions are shown.

Expression A	Expression B
$-191 + 77$	$-191 - (-77)$

With your partner, determine if the expressions are equivalent. Summarize your determination below.

2. Annalise is on a trivia game show. Her score after the first round is -500 points. She misses the first question in the second round, losing an additional 125 points.

Kiersten said she could represent Annalise’s score using subtraction, while Dylan said it could only be represented with addition.

- a. Complete the table with the expression each student was thinking of.

Kiersten’s Subtraction Expression	Dylan’s Addition Expression
_____ - _____	_____ + _____

- b. Kiersten said Annalise’s score was -375 after missing the first question in the second round, but Dylan said it was -625 . Explain which student is correct.

3. Rewrite each subtraction equation as an equivalent addition equation.

Subtraction Equation	$513 - 382 = 131$	$957 - ^{-}136 = 1,093$
Addition Equation		

Addition and subtraction are **inverse operations**, which means they are operations that have the opposite effect of each other.

Understanding this relationship and the existence of additive inverses can be helpful when subtracting integers.



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

4. Complete the statements below using this information.

A subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to _____ and changing the second value to its _____.



6.8.3 Try It: Subtracting Integers

1. The table shows the beginning and ending elevations, in feet (ft), of six different hiking trails. The change in elevation along the trail can be found by subtracting the initial elevation from the final elevation.

$$(\text{final elevation}) - (\text{initial elevation}) = (\text{change in elevation})$$

- a. Write an expression that represents the change in elevation along each hiking trail. Then, write the value that represents the change in elevation.

The first row has been completed as an example.

Initial Elevation (ft)	Final Elevation (ft)	Difference Between Final and Beginning	Change in Elevation (ft)
400	900	$900 - 400$	500
115	68		
294	-121		
-200	610		
-163	-57		
-17	-87		

- b. Explain what a negative change in elevation means in this context.

2. Find each difference.

a. $-664 - 231$

b. $109 - (-115)$



6.8.4 Your Turn!

1. Evaluate each expression.

a. $740 - (-171)$

b. $569 - 143$

c. $-738 - (-212)$

d. $333 - (-204)$

e. $-877 - 814$

f. $-691 - 293$

g. $157 - (-120)$

h. $341 - (-504)$



6.8.5 Wrap-Up: Ticket Out the Door

1. Evaluate each expression.

a. $447 - (-702)$

b. $-834 - (-610)$

c. $-309 - 525$

Unit 6, Lesson 9: Multiplying Integers Using Models



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



Learning Target

- ☐ I can multiply integers using integer chips, number lines, and patterns.